COMP9444 Assignment 1

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Introduction

In this assignment, three neural networks are implemented and trained on the MNIST handwritten digit classification task. The performance of each network is evaluated using the final test accuracy and confusion matrix.

Question 1

Network Structure

The NetLin model is a single-layer linear neural network. Each input image has size 28×28, which is flattened into a vector of length 784. The flattened input is then passed through a linear layer named `self.in_out`, producing 10 output values corresponding to the 10 digit classes.

Result

The final test accuracy of NetLin is 91.53%.

```
Train Epoch: 40 [0/60000 (0%)]  Loss: 0.069893
Train Epoch: 40 [6400/60000 (11%)]      Loss: 0.068813
Train Epoch: 40 [12800/60000 (21%)]     Loss: 0.039252
Train Epoch: 40 [19200/60000 (32%)]     Loss: 0.056744
Train Epoch: 40 [25600/60000 (43%)]     Loss: 0.083029
Train Epoch: 40 [32000/60000 (53%)]     Loss: 0.044769
Train Epoch: 40 [38400/60000 (64%)]     Loss: 0.064232
Train Epoch: 40 [44800/60000 (75%)]     Loss: 0.053419
Train Epoch: 40 [51200/60000 (85%)]     Loss: 0.067410
Train Epoch: 40 [57600/60000 (96%)]     Loss: 0.073966
[[ 961.    0.    1.    2.    0.    4.    9.    2.    1.    0.]
 [   0. 1109.    2.    2.    0.    2.    4.    2.   14.    0.]
 [   7.   14.  913.   15.    9.    6.   12.   14.   36.    6.]
 [   6.    1.   20.  907.    1.   31.    5.   13.   16.   10.]
 [   1.    3.    4.    1.  915.    1.   10.    1.    7.   39.]
 [  10.    3.    1.   31.   10.  778.   19.    5.   26.    9.]
 [  11.    3.    5.    2.    9.   17.  907.    2.    2.    0.]
 [   3.   11.   21.    6.    9.    2.    1.  939.    2.   34.]
 [  12.   15.    7.   16.   15.   42.   13.   13.  830.   11.]
 [   9.    9.    2.   11.   39.   10.    0.   26.    9.  894.]]

Test set: Average loss: 0.0010, Accuracy: 9153/10000 (92%)
```

Figure 1: Final accuracy and confusion matrix for NetLin.

Question 2

Confusion Matrix Analysis

It can be observed from the confusion matrix that there are more mistakes in identifying 4s and 9s; both of these digits are often mistaken for each other. Moreover, 8s are often mistaken for 5s, 2s are often mistaken for 8s, and 3s and 5s are often mistaken for each other. It is very probable that these mistakes arise due to their similar structure.

Question 3

Weight Visualization

The following images show the learned input-to-output weights of NetLin for digit classes 0 to 9.

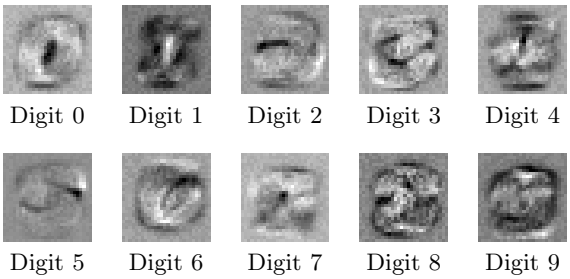


Figure 2: Weight visualisations of NetLin for digit classes 0 to 9.

Question 4

Bright regions correspond to positive weights, meaning that pixels in these regions increase the score for the corresponding digit. Dark regions correspond to negative weights, meaning that pixels in these regions reduce the score and help reject similar digit classes.

- **Digit 0:** The bright regions are mainly around the outer circular boundary, while the dark regions appear near the centre. This helps distinguish digit 0 from digit 8, which usually contains additional strokes through the centre.
- **Digit 1:** The bright regions are concentrated along the central vertical stroke, while the dark regions suppress unnecessary strokes on both sides. This helps distinguish digit 1 from digits 4 and 7.
- **Digit 2:** The bright regions highlight the upper curve and lower horizontal stroke. The dark regions discourage pixels where a closed loop would appear. This helps distinguish digit 2 from digit 8.

- **Digit 3:** The bright regions emphasise the upper and lower curved strokes, while the dark regions on the left discourage a strong vertical stroke. This helps distinguish digit 3 from digit 5.
- **Digit 4:** The bright regions correspond to the vertical and diagonal strokes, while the dark regions appear around the lower-left area. This helps distinguish digit 4 from digit 9.
- **Digit 5:** The bright regions emphasise the upper horizontal stroke and lower curve, while the dark regions suppress pixels near the upper-right region. This helps distinguish digit 5 from digit 3.
- **Digit 6:** The bright regions highlight the lower loop, while the dark regions reduce the influence of strokes in the upper-right area. This helps distinguish digit 6 from digit 0.
- **Digit 7:** The bright regions correspond to the top horizontal stroke and diagonal stroke, while the dark regions discourage pixels in the lower-left area. This helps distinguish digit 7 from digit 1.
- **Digit 8:** The bright regions highlight both loops, while the dark regions suppress pixels outside the loops. This helps distinguish digit 8 from digit 0.
- **Digit 9:** The bright regions emphasise the upper loop and right vertical stroke, while the dark regions suppress pixels in the lower-left region. This helps distinguish digit 9 from digit 4.

Question 5

Network Structure

The NetSkip model extends NetLin by adding the hidden layer with eight nodes and the skip connection. In the beginning, the input image is flattened to 784 dimensions. The flattened input is fed into the hidden layer as well as the output layer at the same time. The hidden layer uses ReLU activation function and its output is further forwarded to the output layer. The final output is generated by combining the input-output pathway with the hidden-output pathway.

Result

The final test accuracy of NetSkip is 93.95%.

```
Test set: Average loss: 0.0008, Accuracy: 9383/10000 (94%)

Train Epoch: 40 [0/60000 (0%)]  Loss: 0.065653
Train Epoch: 40 [6400/60000 (11%)]      Loss: 0.052717
Train Epoch: 40 [12800/60000 (21%)]     Loss: 0.073004
Train Epoch: 40 [19200/60000 (32%)]     Loss: 0.064966
Train Epoch: 40 [25600/60000 (43%)]     Loss: 0.050888
Train Epoch: 40 [32000/60000 (53%)]     Loss: 0.051810
Train Epoch: 40 [38400/60000 (64%)]     Loss: 0.043523
Train Epoch: 40 [44800/60000 (75%)]     Loss: 0.052117
Train Epoch: 40 [51200/60000 (85%)]     Loss: 0.041816
Train Epoch: 40 [57600/60000 (96%)]     Loss: 0.070789
[[ 963.    0.    1.    2.    0.    3.    7.    1.    3.    0.]
 [   0. 1120.    2.    1.    0.    2.    4.    0.    6.    0.]
 [  12.    6.  952.   10.   12.    0.    6.   14.   18.    2.]
 [   1.    3.   19.  937.    1.   21.    3.   12.    9.    4.]
 [   1.    3.    8.    0.  936.    0.    7.    2.    3.   22.]
 [  14.    3.    1.   24.    4.  805.   10.    6.   17.    8.]
 [   9.    3.    2.    1.    8.   16.  915.    1.    3.    0.]
 [   3.   11.   18.    7.    5.    0.    2.  957.    1.   24.]
 [   9.    5.    4.   25.    9.   15.   12.    8.  882.    5.]
 [   9.    7.    2.   16.   33.    5.    1.    4.    4.  928.]]

Test set: Average loss: 0.0008, Accuracy: 9395/10000 (94%)
```

Figure 3: Final accuracy and confusion matrix for NetSkip.

Question 6

Weight Visualization

The following images show the learned input-to-hidden weights for the eight hidden nodes of NetSkip, together with the hidden-to-output weight matrix.

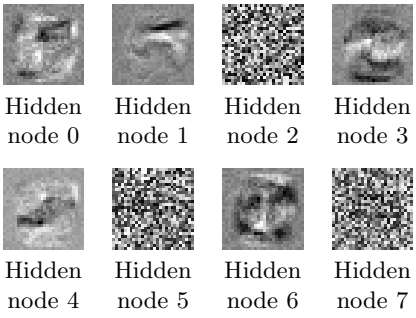


Figure 4: Input-to-hidden weight visualisations for the eight hidden nodes.

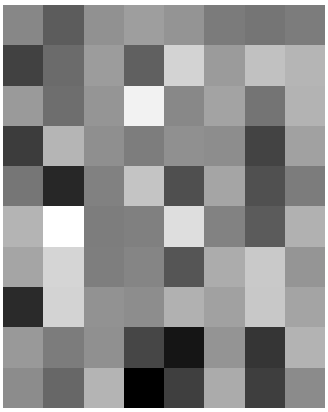


Figure 5: Hidden-to-output weight matrix of NetSkip.

Question 7

Some hidden nodes seem to be used quite poorly. In particular, weight images for hidden nodes 2, 5, and 7 look like noise rather than something meaningful like digit features. Moreover, columns of these hidden nodes in `hid_out.png` look very much like uniform grey color. Therefore, we can say that these hidden nodes hardly influence the final classification result.

Hidden nodes 0, 1, 3, 4, and 6 learn meaningful visual features. In `hid_out.png`, rows correspond to output classes 0-9 and columns correspond to hidden nodes 0-7. Hidden node 0 learns broad curved stroke features that could be useful for classifying digits like 0, 6, 8, and 9 and distinguishing them from linear digits. Hidden node 1 responds to horizontal strokes and could be useful for separating digits like 2 and 5. Hidden node 3 learns curved features and could help to separate digits like 3 and 5. Hidden node 4 responds to slanted or low stroke features and could be used to distinguish digits like 2, 4, and 7. Hidden node 6 has a circular feature and thus could be useful for distinguishing looped digits like 0, 6, 8, and 9.

In general, bright regions denote positive weights that lead to activation when corresponding stroke is detected and dark regions denote negative weights that inhibit such activation. Contrast bright and dark regions in the corresponding columns in `hid_out.png` show that different hidden nodes contribute differently to different digit classes. As a result, NetSkip can distinguish visually similar digits better than NetLin.

Question 8

Network Structure

The NetConv model is made up of two convolutional layers followed by two fully connected layers. The first convolutional layer uses 4 filters of size 8×8 with stride 2, followed by a ReLU activation function. The second convolutional layer uses 16 filters of size 5×5 with stride 2, followed by another ReLU activation. These feature maps are flattened into 256 dimensional vectors that pass through the next fully-connected layer, which has 100 hidden units and ReLU activations.

Result

The final test accuracy of NetConv is 98.65%.

```
Train Epoch: 40 [0/60000 (0%)]  Loss: 0.002679
Train Epoch: 40 [6400/60000 (11%)]      Loss: 0.009224
Train Epoch: 40 [12800/60000 (21%)]     Loss: 0.000918
Train Epoch: 40 [19200/60000 (32%)]     Loss: 0.001636
Train Epoch: 40 [25600/60000 (43%)]     Loss: 0.002080
Train Epoch: 40 [32000/60000 (53%)]     Loss: 0.000914
Train Epoch: 40 [38400/60000 (64%)]     Loss: 0.002393
Train Epoch: 40 [44800/60000 (75%)]     Loss: 0.003122
Train Epoch: 40 [51200/60000 (85%)]     Loss: 0.000545
Train Epoch: 40 [57600/60000 (96%)]     Loss: 0.004480
[[ 971.    0.    0.    0.    0.    0.    5.    1.    3.    0.]
 [   0. 1128.    0.    1.    0.    1.    1.    1.    2.    1.]
 [   0.    1. 1018.    1.    1.    0.    2.    4.    5.    0.]
 [   0.    0.    2.  997.    0.    2.    0.    2.    4.    3.]
 [   1.    0.    3.    0.  967.    0.    2.    1.    0.    8.]
 [   1.    1.    0.    7.    0.  872.    4.    1.    3.    3.]
 [   1.    1.    0.    0.    1.    3.  950.    0.    2.    0.]
 [   0.    2.    4.    4.    0.    0.    0. 1010.    1.    7.]
 [   0.    0.    2.    4.    0.    2.    1.    1.  961.    3.]
 [   3.    0.    0.    1.    6.    2.    0.    3.    3.  991.]]

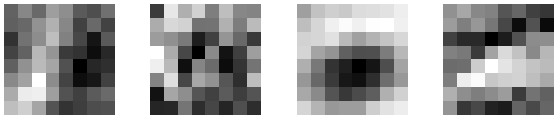
Test set: Average loss: 0.0002, Accuracy: 9865/10000 (99%)
```

Figure 6: Final accuracy and confusion matrix for NetConv.

Question 9

Filter Visualization

The following images show the four learned filters from the first convolutional layer of NetConv.



Filter 0 Filter 1 Filter 2 Filter 3

Figure 7: Visualisations of the four filters in the first convolutional layer.

Question 10

In the first convolutional layer, the filters used by the network are trained in detecting different kinds of low-level features such as edges, strokes, and curves. Positive weights represent the brighter parts which help in boosting the response of the filter while negative weights indicate the darker parts which help in decreasing the response.

- **Filter 0:** It seems like this filter detects diagonal edges. The bright area at the bottom left corner and the dark area at the right show that the response is strong towards diagonally changing pixel intensities. This could be triggered by the diagonal strokes of digits such as 7 and 2, helping distinguish them from round digits like 0 and 8.
- **Filter 1:** The activation is noticed to be selective for diagonal edges and corner-like characteristics. The brighter spots on the upper left and darker areas near the center and lower right imply that there is an appreciation for the variations in the local stroke orientation. This could be triggered by digits like 4 and 7 to distinguish them from rounder digits like 0 and 6.
- **Filter 2:** The filter seems to be identifying shapes that are looped or curved. The presence of bright spots around the outer boundary and the presence of a dark spot in the center implies that the filter reacts to closed shapes having an internal hollow. It might be highly triggered by the digits like 0, 6, 8, and 9, which will help distinguish them from other less closed shapes like 1 and 7.
- **Filter 3:** The filter appears to detect junctions or intersection of the strokes. The areas that appear bright in the center and dark around the top edge suggest a high level of activation in detecting crossings or corners. This is because it may be stimulated by digits like 4, 8, and 9, thus helping in differentiating them from other simpler digits like 1 and 0.